



Agenda

- What is Angular?
- What are the alternatives to Angular?
- Single Page Application
- Getting started with angular
- Angular building blocks: components.
- Templates and styles
- Data Binding





What is Angular ?

- Angular is a **JavaScript framework** that allows you to create single-page applications. It's only one HTML file and a bunch of JavaScript code we got from the server that changes the content of this HTML.
- Angular is using **Typescript**, which is a superset of Javascript and compiles to Javascript.
- Current stable version is **v20**
- Angular was developed by Google team
- Angular docs : <https://angular.dev/>



What is Angular ?

- A component-based framework for building scalable web applications
- A collection of well-integrated libraries that cover a wide variety of features, including routing, forms management, client-server communication, and more
- A suite of developer tools to help you develop, build, test, and update your code



Angular Alternates

Stack Overflow Survey:

<https://survey.stackoverflow.co/2024/technology/#1-web-frameworks-and-technologies>



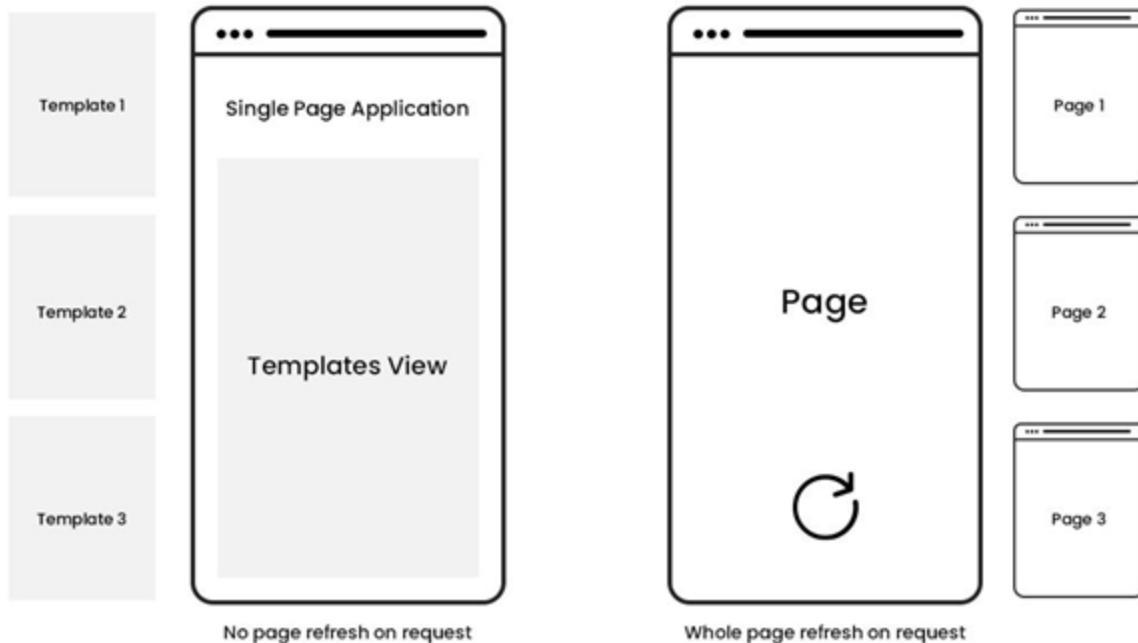


Single page application

- Single Page Applications (SPAs) all the HTML generation happens in the browser. The server only returns one basic HTML page for all incoming requests (no matter the URL).
- But that single HTML page contains a lot of JavaScript code (typically outsourced into separate files) which is responsible for changing the HTML code (technically, the DOM).
- Single-page application examples : gmail,facebook,netflix ...etc



Single-page application





Angular : Getting Started

- Install npm (node package manager) :

<https://nodejs.org/en/>

What is NPM ?

Node Package Manager (NPM) is a command line tool that installs, updates or uninstalls Node.js packages in your application. It is also an online repository for open-source Node.js packages. The node community around the world creates useful modules and publishes them as packages in this repository.



Angular : Getting Started

- Install Angular CLI (command line interface) globally :

npm install -g @angular/cli

What is CLI ?

The Angular CLI is a command-line interface tool that you use to initialize, develop, scaffold, and maintain Angular applications directly from a command shell.

Then try to run **ng version** to make sure that angular/cli installed.



Angular : Getting Started

To create your first angular app , run the following command in CMD :

- **ng new project-name** (create angular app)
- **cd ./project-name** (enter project folder)
- **ng serve -o** (run application and -o (--open) refers to open application automatically in browser)



Angular : Getting Started

Let's discover and have a deep
look at our angular app structure

<https://angular.dev/reference/configs/file-structure>

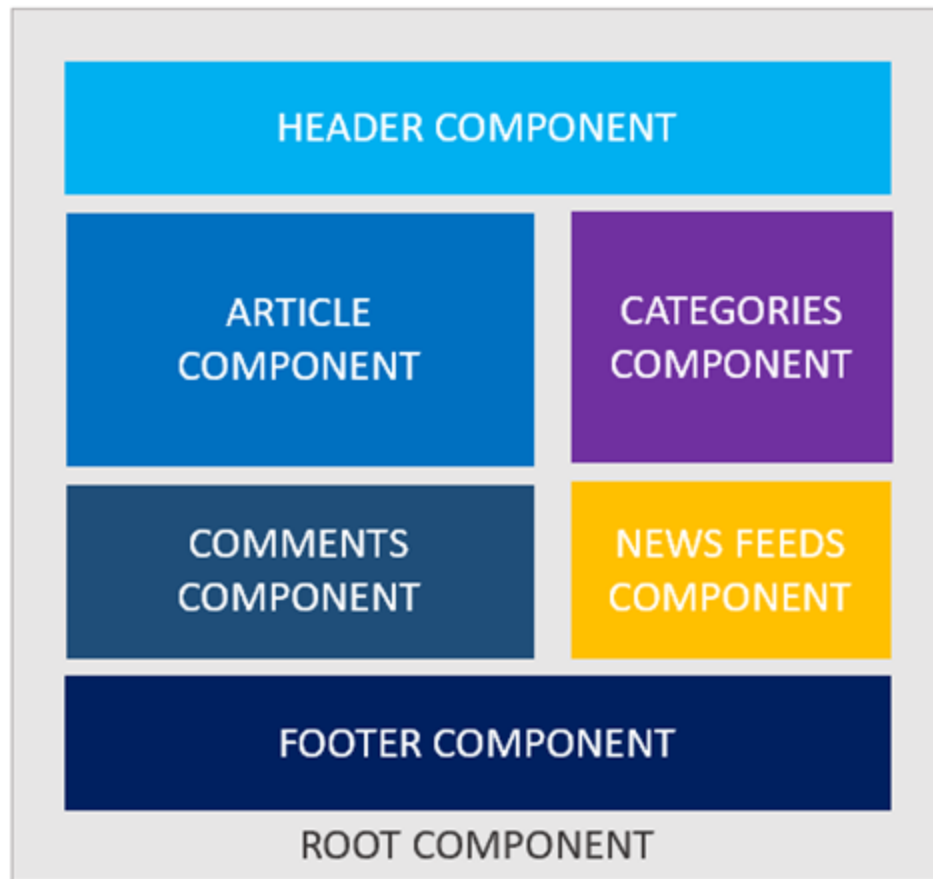




component



Components





Components

Components are the main building block for Angular applications ,
Components are composable, we can build larger Components from smaller ones. Each component consists of:

- An HTML template that declares what renders on the page
- A Typescript class that defines behavior
- A CSS selector that defines how the component is used in a template
- Spec file for testing



Standalone components

Components, directives, and pipes can now be marked as standalone: true. Angular classes marked as standalone do not need to be declared in an NgModule

Standalone components specify their dependencies directly instead of getting them through NgModules

<https://angular.dev/guide/components/importing>



Components

- Create new component : **ng generate component navbar**
- Discover new generated component class.
- @Component decorator: decorator that marks a class as an Angular component and provides configuration metadata that determines how the component should be processed, instantiated, and used at runtime.
- External template and styles





Reusable components

- To use component in any other place you can use it by selector name mentioned in `@component` between HTML tags.
- Create once , use multiple times.
- Syntax : `<app-navbar></app-navbar>`
- Add component to imports as it's a standalone component



ANGULAR

Data Bindings

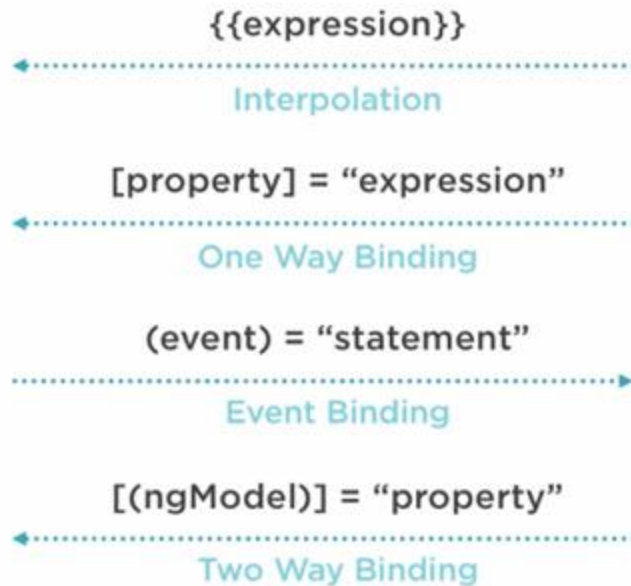


Data binding

<https://angular.dev/guide/templates>



DOM



Component



Data binding

- **String Interpolation**

Allows you to incorporate dynamic string values into your HTML templates

used like this `{{expression}}`

- **Property Binding**

Property binding in Angular helps you set values for properties of HTML elements or directives.

Example : `<img [src]="itemImageUrl">`



Data binding

- **Event Binding**

Event binding allows you to listen for and respond to user actions such as keystrokes, mouse movements, clicks, and touches.

Example : `<button (click)="onSave()">Save</button>`

- **Two way binding**

Two-way binding combines property binding with event binding for example to two way binding `[(ngModel)]`

Note : need to import `import { FormsModule } from '@angular/forms'` for two way binding in forms in app module to work

Thanks!





Portfolio

Replace code found in app.component.html and start create your portfolio page using Bootstrap.

Portfolio will contain the following sections :

- Hero section (name and job title).
- Bio and about me (education and experiences) section with button to download CV .
- Skills section with progress bar for each skill
- Portfolio and projects section
- Footer contains contact us section with email and social media links (facebook , github , linkedin) [Will use fontawesome - **Bonus**]

Each section is a separate component.

